

Mahima Chotiya

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Education

Indian Institute of Technology, Delhi (IITD)

Class of 2028

B.Tech in Mechanical Engineering

Current GPA: 7.3/10

- **Selected Coursework:** Mathematical Physics, Electromagnetic Waves, Quantum Mechanics*, Calculus, Linear Algebra & Differential Equations, Engineering Drawing, Computer Science

Matrix High School

Class of 2024

Secondary Education: 94.8% Higher Secondary Education: 87.2%

- **JEE Main 2024:** Secured **99.84** percentile with **AIR 327(EWS)** among **1.6+** million candidates
- **JEE Advanced 2024:** Secured All India Rank **1114(EWS)** among **200,000+** candidates
- **International Mathematics Olympiad:** Class Rank **1**, State Rank **13**, and Olympiad Rank **718**.
- **Mathematics Olympiads:** Qualified **IOQM** (Stage 1) and **RMO** (Stage 2), and selected for **INMO**.
- **Kotak Kanya Scholar:** Awarded to only **303** meritorious students across India

Internship

University of South Wales *Marketing Lab Intern*

[May'25–Jul'25]

- Developed and evaluated **three anime/manga** content pipelines to study real-world **GenAI** integration in the entertainment industry.
- Reduced content production time by **40–60** workflows and analyzed audience engagement, ratings, and feedback to assess consumer acceptance of **AI-generated** content.
- *Launched* and *tested AI-generated* digital products and branding assets across online platforms, supporting marketing strategy experiments through performance and audience response analysis.

Projects

Masheasy

[Mar'25–May'25]

Group led by Prof. Sagar Sarkar (Dept. of Mechanical Engineering, IITD)

- Developed Mash Easy, a **handmade** product offering a **low-cost** alternative to expensive mashers.
- **Identified affordability** as a key barrier and developed a low-cost, practical, user focused solution.

Frequency Divider

[Oct'24–Nov'24]

Group led by Prof. Mummadi Veerachary (Dept. of Electrical Engineering, IITD)

- Designed a Frequency Divider circuit using **flip-flops** and **counters** to reduce input frequency.
- Applied digital logic to generate **lower-frequency signals** for timing, clocking, and signal processing.

Graph-Based Friend Recommendation System (Java)

[Mar'25–May'25]

- Developed a Java-based recommender leveraging **OOP**, **HashMap** **Stream API** for dynamic adjacency-list graphs.
- **Applied Common Neighbors, Jaccard, Adamic-Adar** graph algorithms random walks for Node2Vec embeddings.
- Scaled to **10K** users/**50K** edges; **top-N** in $O(N \log N)$, **1s/query**, with **3× higher precision@10** vs. baseline(**n = 1K**).

Leadership and Activities

National Service Scheme (NSS)

[Aug'24–Nov'24]

- **Bachpan Project (IIT Delhi):** Taught children of campus through interactive learning sessions
- **Blood Donation Camp:** Supported execution of the camp and participant management
- **Old Age Home Visit:** Engaged in conversations and recreational activities to reduce loneliness
- **Cleanliness Drive:** Worked with team members to improve living surroundings

IIT Delhi Sports Activity

[Jan'25–Jan'26]

Vice-Captain, Badminton Team

- Played a key leadership role in securing multiple match victories in the General Championship.
- Motivated team members and maintained high performance under pressure.

Skills / Interests

Technical: C, C++, Python, Java, SQL, JavaScript, CAD, SolidWorks, Data Structures & Algorithms

Languages: English, Hindi